SYSDESIGN REFERENCE

Design a Ride-Sharing Dispatch

Requirements-to-architecture reference walkthrough.

Match supply and demand using fresh geospatial state under rapid change.

VERSION 1.0 REVIEW

REVIEWED THROUGH 2026-07-18

STATUS HUMAN REVIEW REQUIRED

SYSDESIGN / RIDE-SHARING-DISPATCH

SYSDESIGN

Design a Ride-Sharing Dispatch

CORE IDEA

Match supply and demand using fresh geospatial state under rapid change.

Reference architecture decision chain

01	02	03	04
Location Indexing	Candidate Search	Scoring	Reservation

Decisions to defend

+ Location indexing
+ candidate search
+ scoring
+ reservation
+ consistency
+ and regional partitioning.

Failure modes to test

+ Double assignment
+ stale locations
+ surge spikes
+ and city-level outages.

Interview flow

- [ ] Clarify users, critical paths, and non-goals
- [ ] Estimate only scale that changes a decision
- [ ] Define API, data model, and source of truth
- [ ] Draw synchronous and asynchronous paths
- [ ] Pressure-test timeout, retry, overload, and recovery
- [ ] Close with tradeoffs and one alternative

Deep-dive drill

Design a Ride-Sharing Dispatch

Probe	What a strong answer makes explicit
How does the design behave during a regional dependency failure?	Requirement, decision, cost, failure behavior, and observable metric.
Which metric reveals the first bottleneck?	Requirement, decision, cost, failure behavior, and observable metric.
What consistency does the user actually observe?	Requirement, decision, cost, failure behavior, and observable metric.
How would the design change at ten times the traffic?	Requirement, decision, cost, failure behavior, and observable metric.

Data questions

+ Partition key
+ Versioning
+ Uniqueness
+ Retention
+ Reconciliation

Operations questions

+ SLO
+ Backlog
+ Capacity
+ Alert
+ Failover test

DESIGN DISCIPLINE

For every box: which requirement justifies it, what cost does it introduce, and what happens when it fails?

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